

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit 5 : Session 10 : SLO 2 | LDM/STM and RISC vs CISC

I. Multiple Choice

1. C) STM Store Multiple saves a block of registers to memory.
2. B) RISC RISC architectures typically use fixed-length instructions.
3. B) Write-back '!' updates the base register after the transfer.

II. Fill in the Blanks

1. Load LDM = Load Multiple registers.
2. variable-length CISC uses variable instruction size.
3. stack STMFD (Store Multiple Full Descending) is the standard stack-push operation.

III. True or False

1. TRUE.
2. FALSE CISC supports more addressing modes than RISC.
3. TRUE write-back (!) updates the base register automatically.

IV. Match the Following

- A) RISC 3) Simple fixed-length instructions.
B) CISC 2) Complex instructions.
C) LDM 1) Loads multiple registers.
D) STM 4) Stores multiple registers.

V. Numerical Answers

1. 12 bytes 3 registers 4 bytes each.
2. 24 bits 32 8 = 24 bits left for operands.
3. STMFD SP!, {R1R3} with SP = 0x1000: a full-descending push of 3 words decreases SP by 12 new SP = 0x0FF4 (data stored at 0x0FF40x0FFF).